

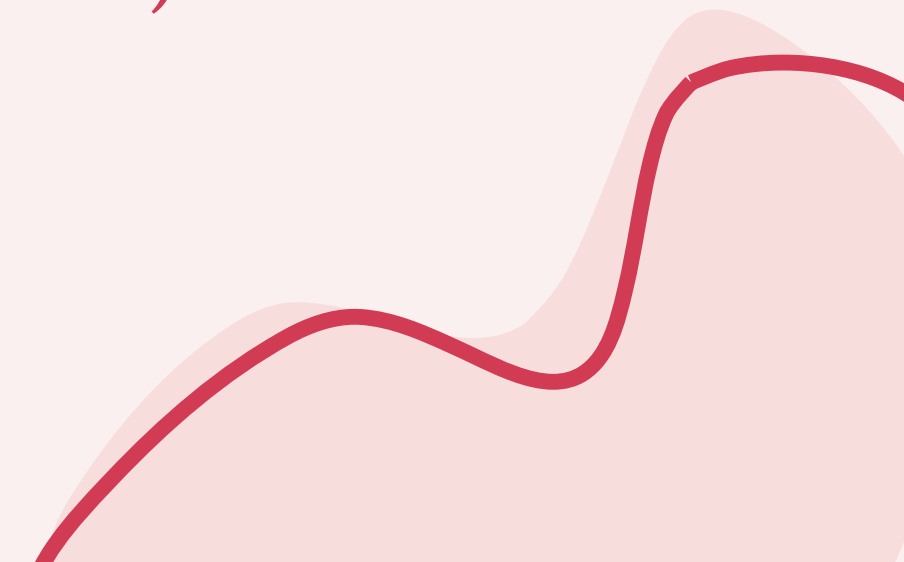
Predicting Readmission Rate for Diabetic Patients Using ML

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Motivation

Problem: Hospital readmissions among diabetic patients remain **common** and **costly**

Policy: CMS Hospital Readmissions Reduction Program: financially penalizes hospitals with high 30-day readmission rates

Goal: Understanding which clinical, demographic, and treatment-related factors drive readmissions can guide patient outcomes and hospital resource allocation

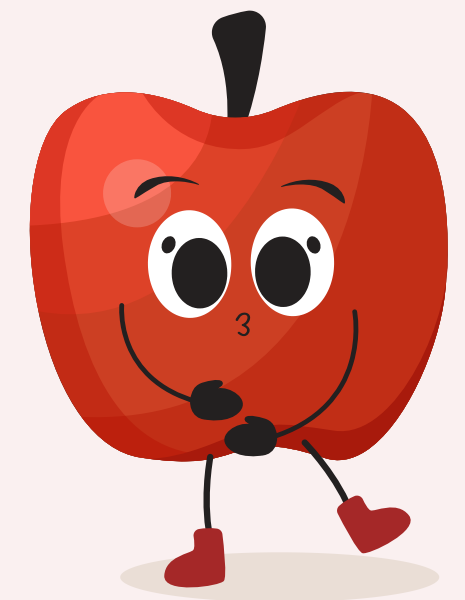


Research Questions

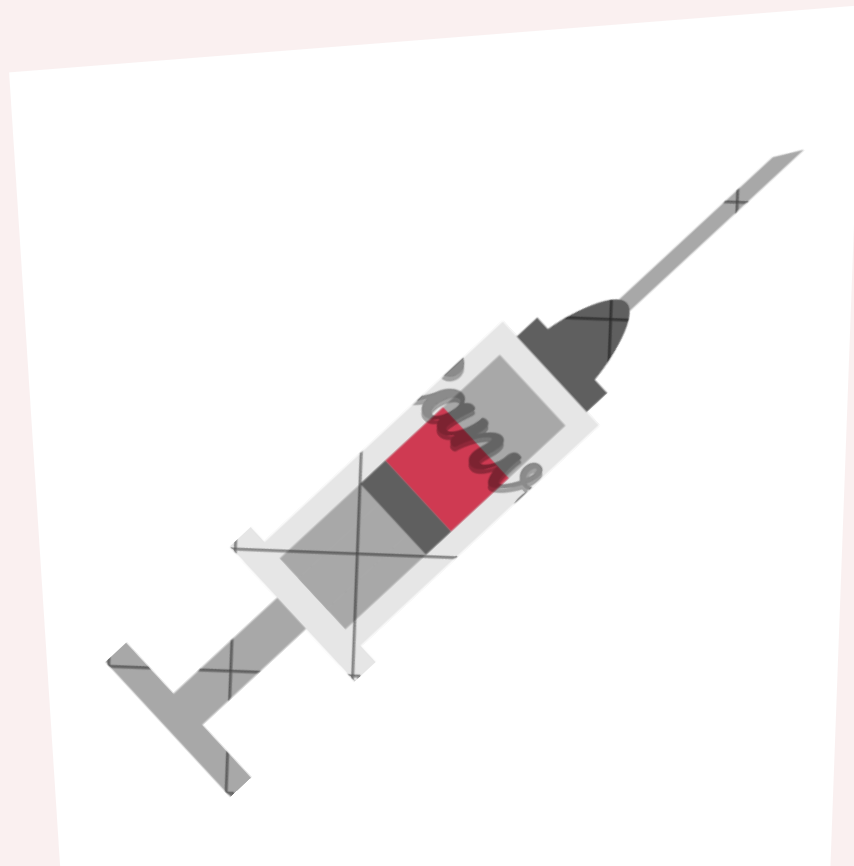
Primary Research Question: *How accurately can ML models predict hospital readmissions among diabetic patients?*

Secondary Research Questions:

- Which features (demographic and clinical) most strongly influence readmissions?
- Which binary ML classifier provides the best predictive performance for identifying readmissions?



Related Work



Developing non-linear
models for clinical risk
predictions



Feature engineering
for treatment
intensity and
change variables

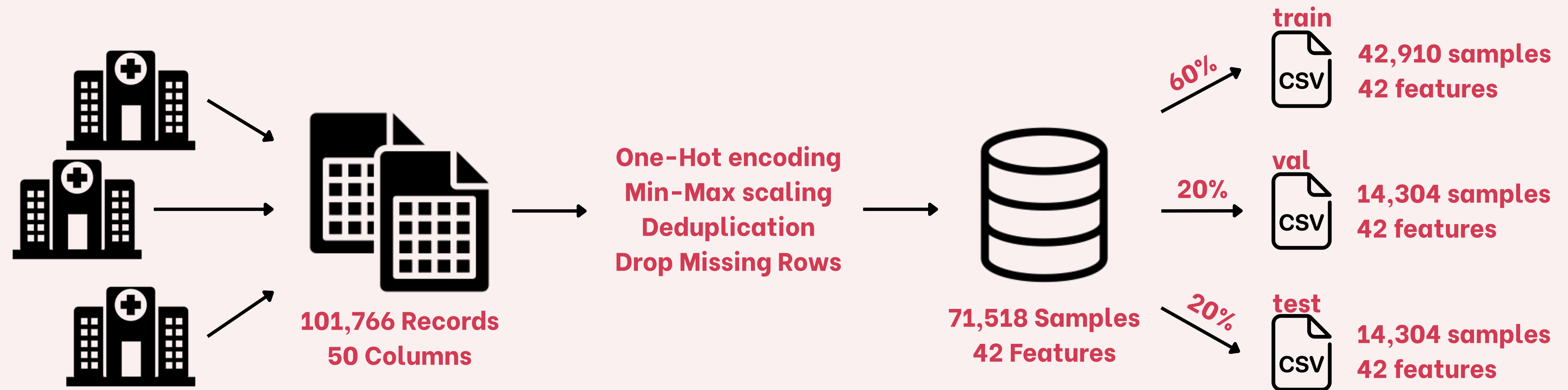


Benchmarking
multiple model
classes

Data & Preprocessing

UCI Diabetes Dataset (1998 and 2008):

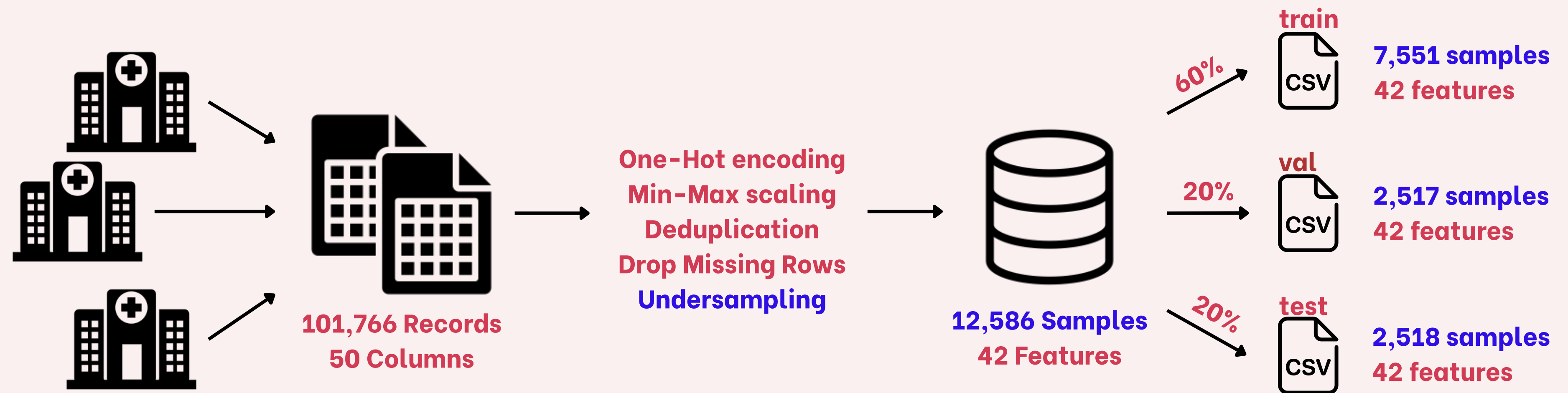
100k+ patient encounters from 130 different hospitals

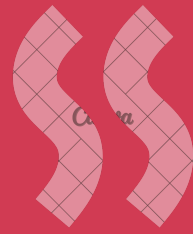


Data & Preprocessing

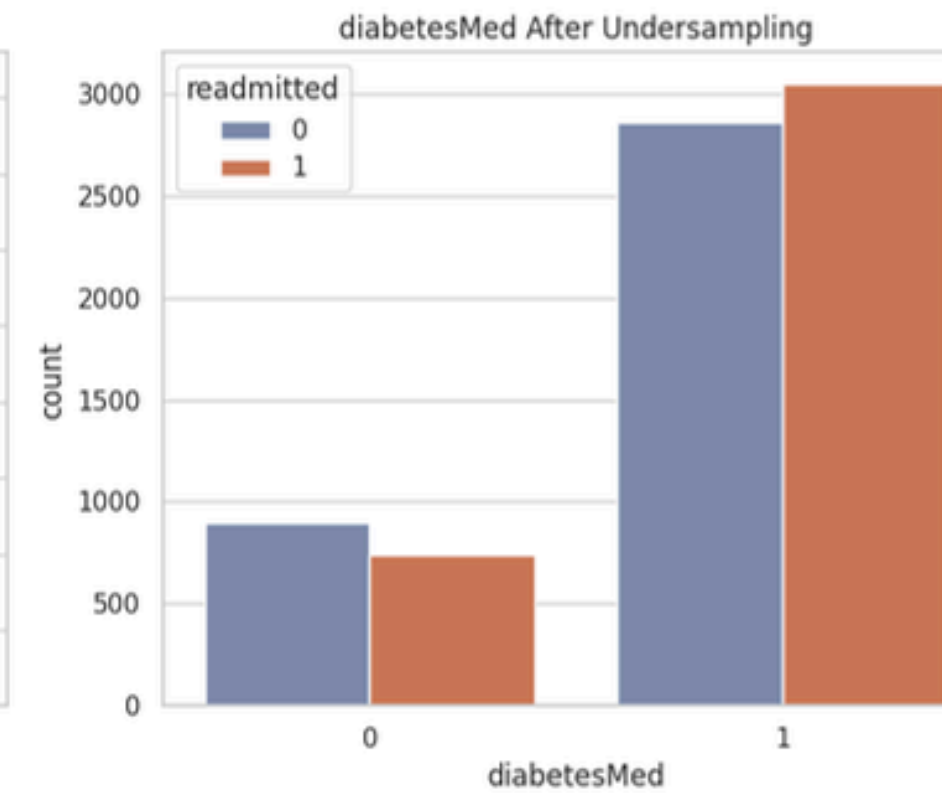
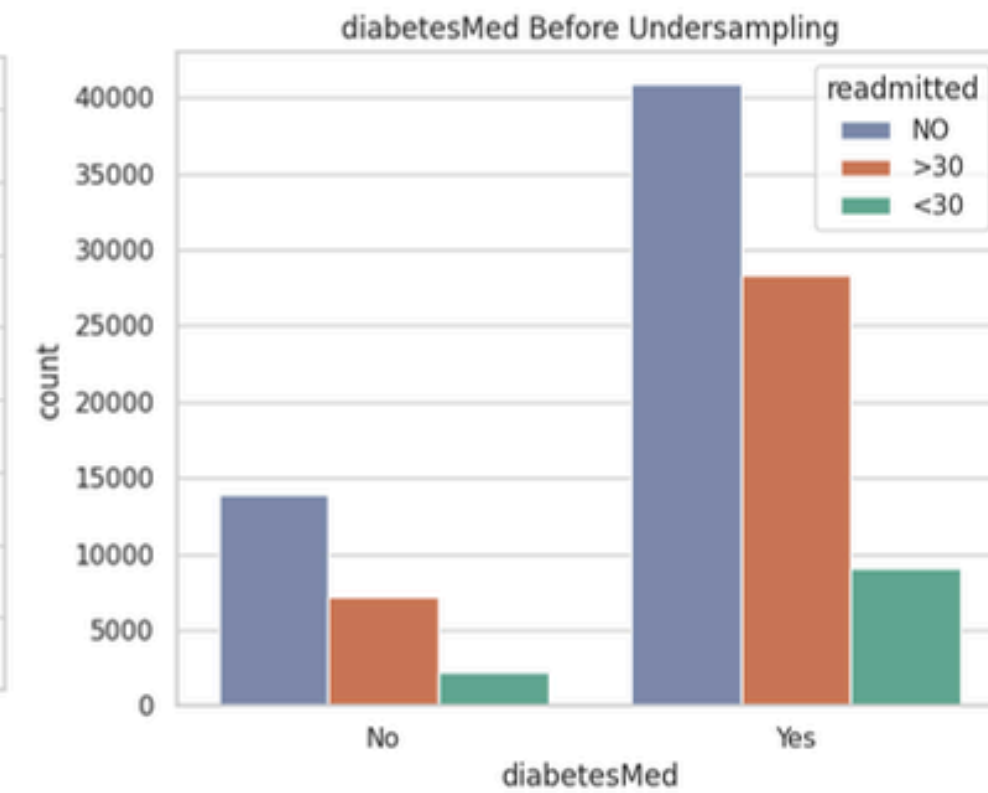
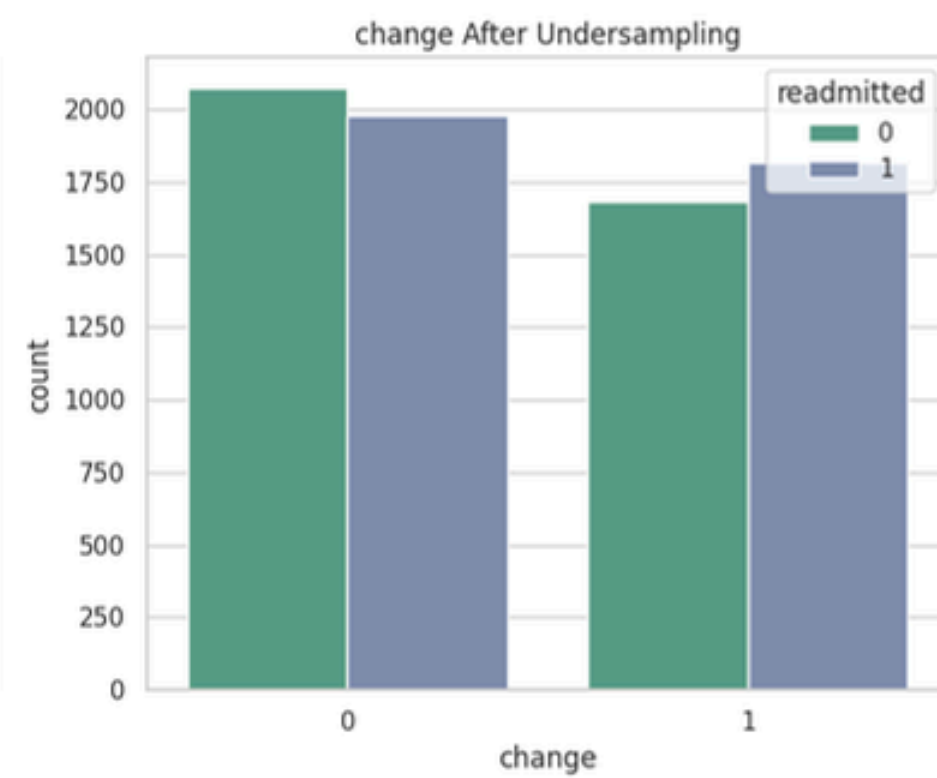
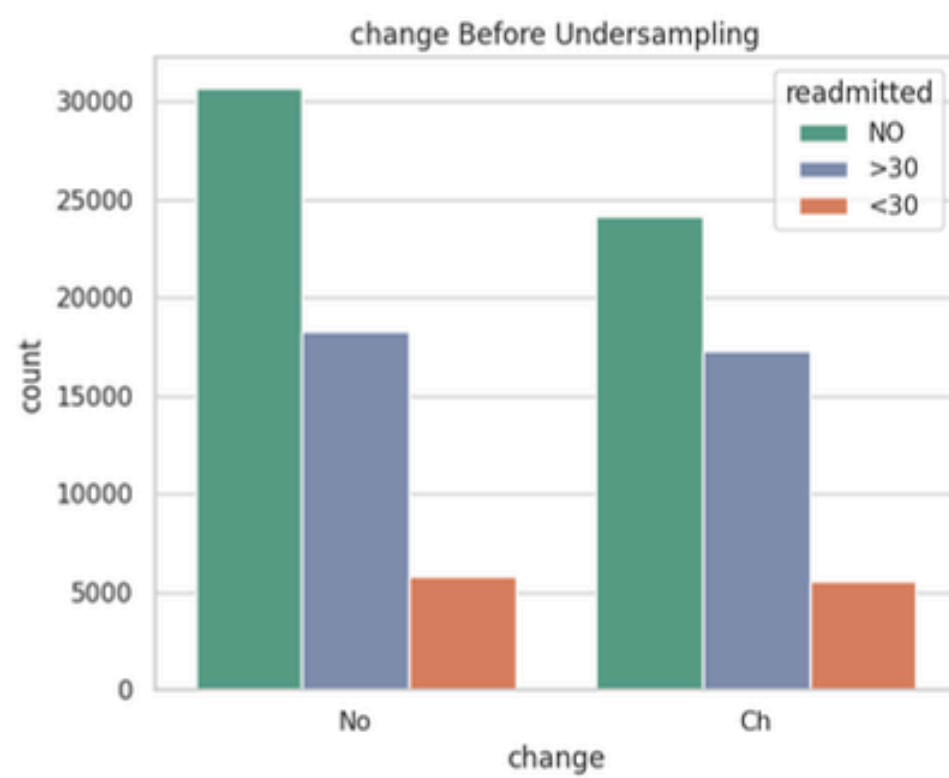
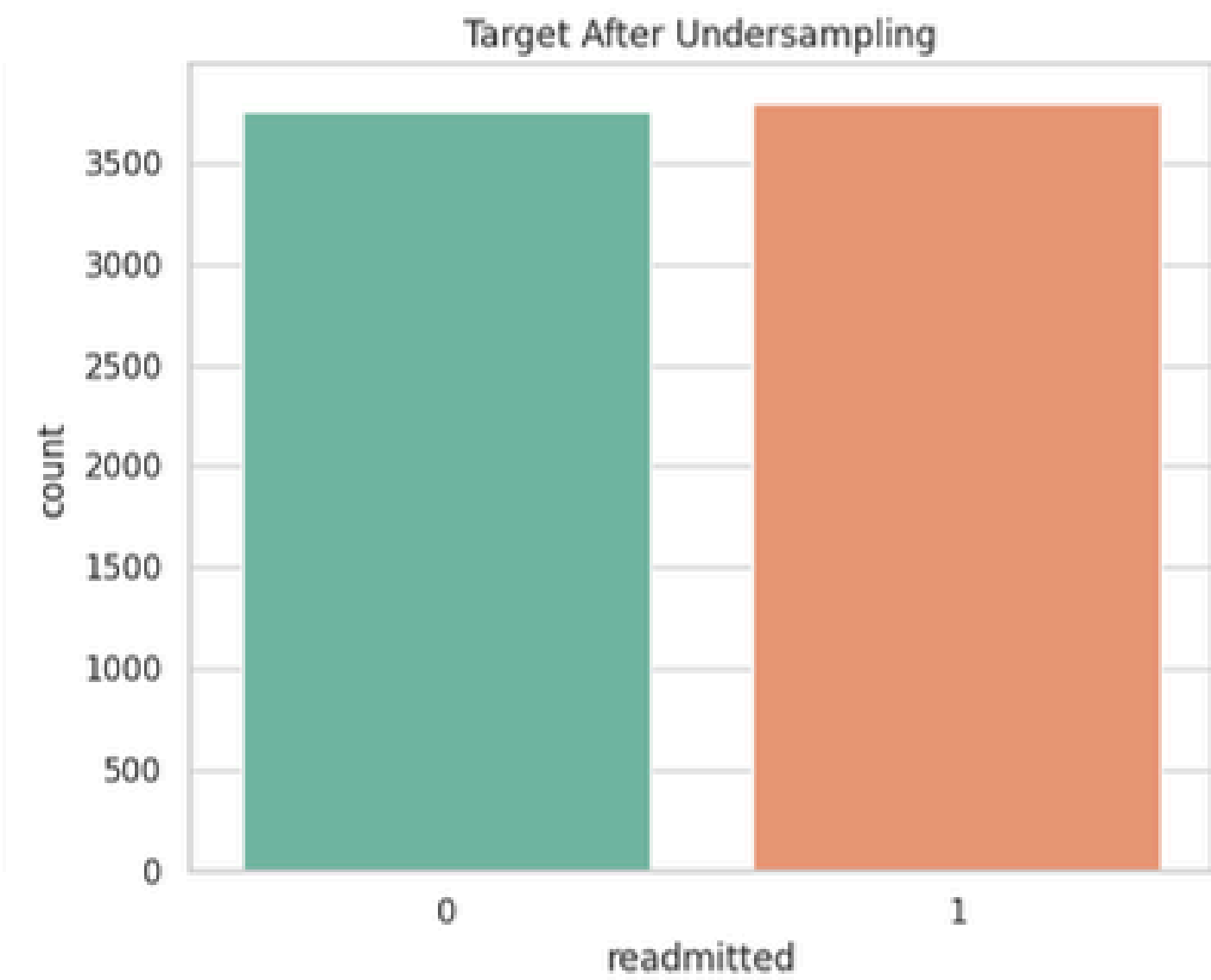
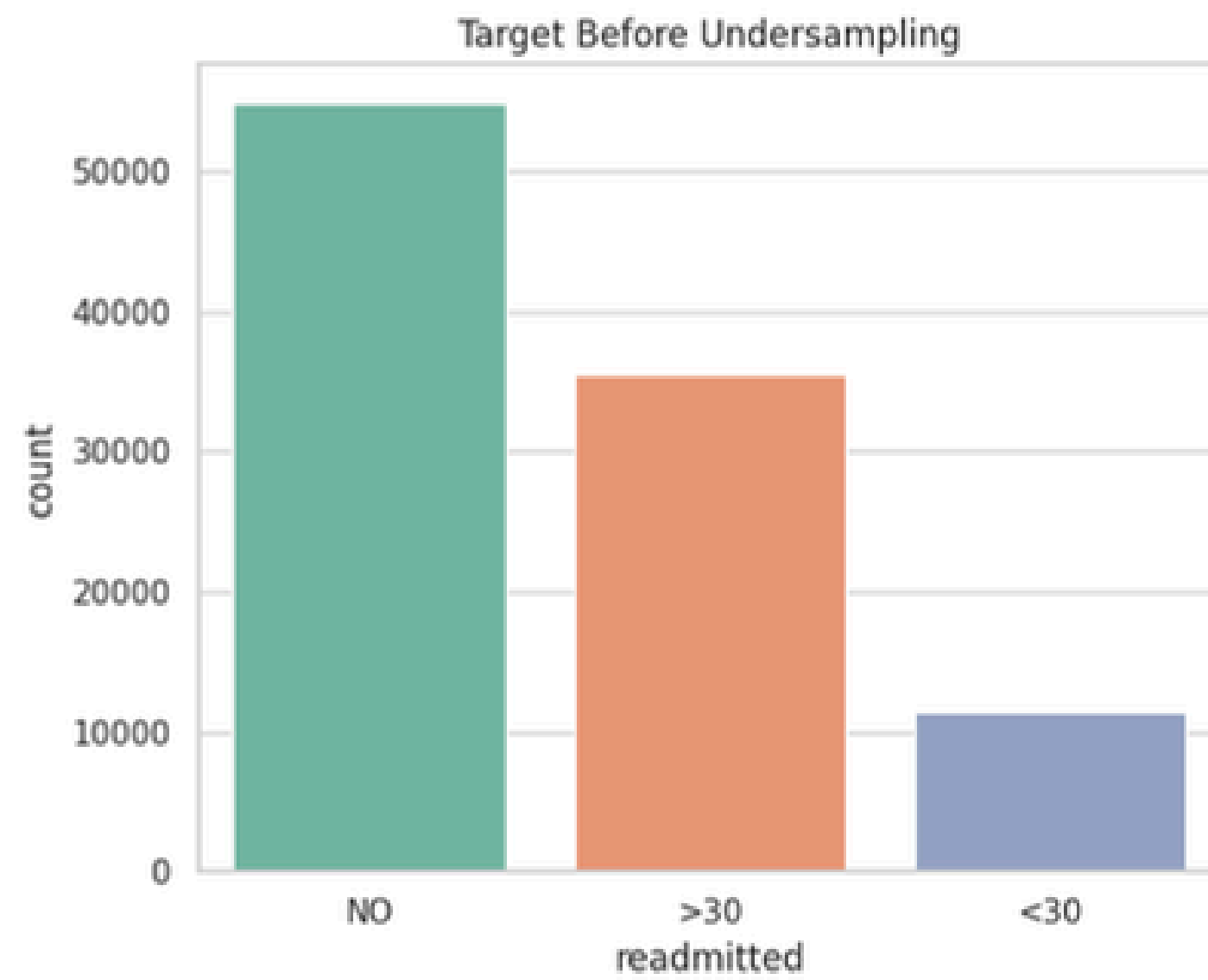
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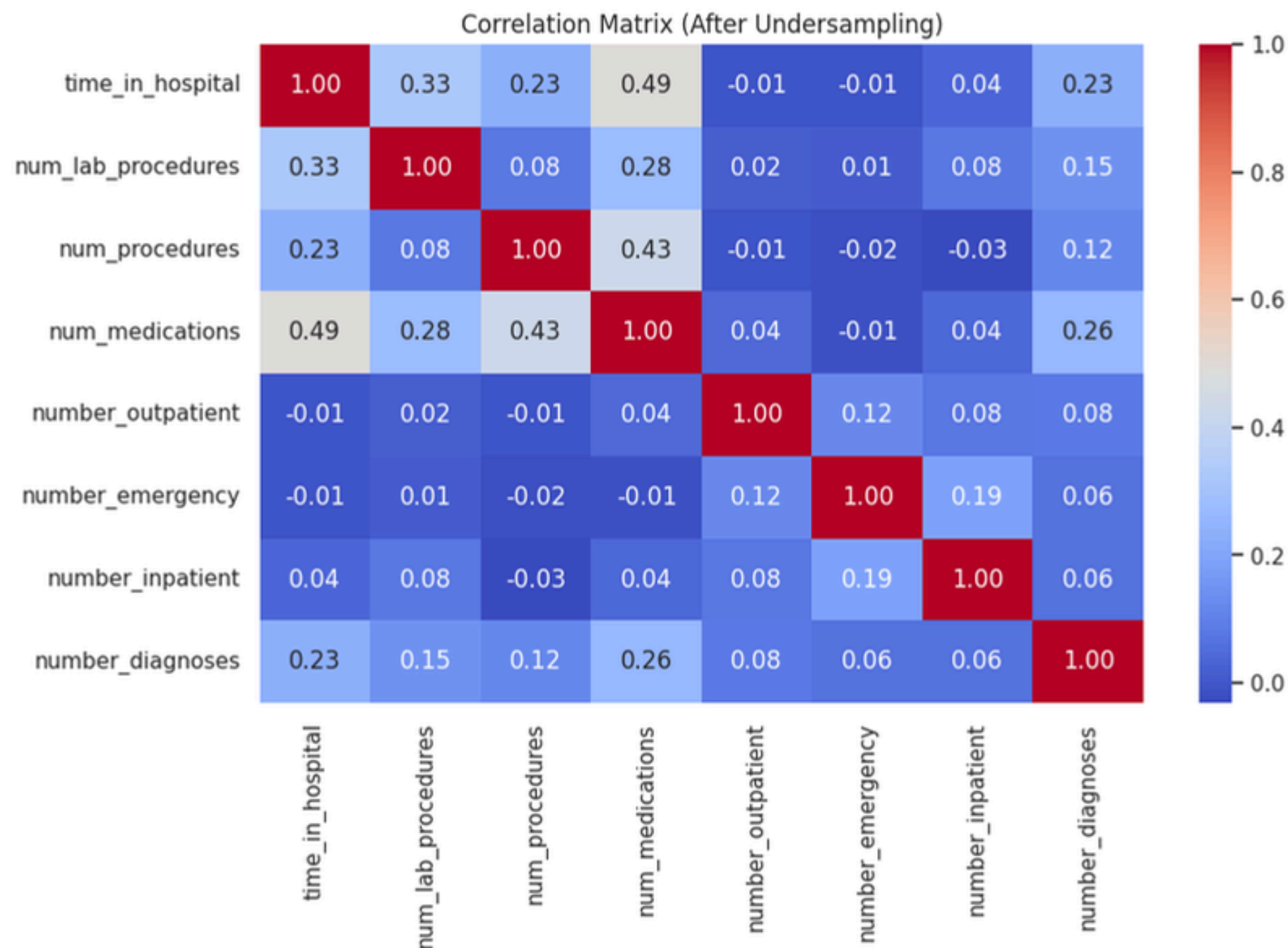
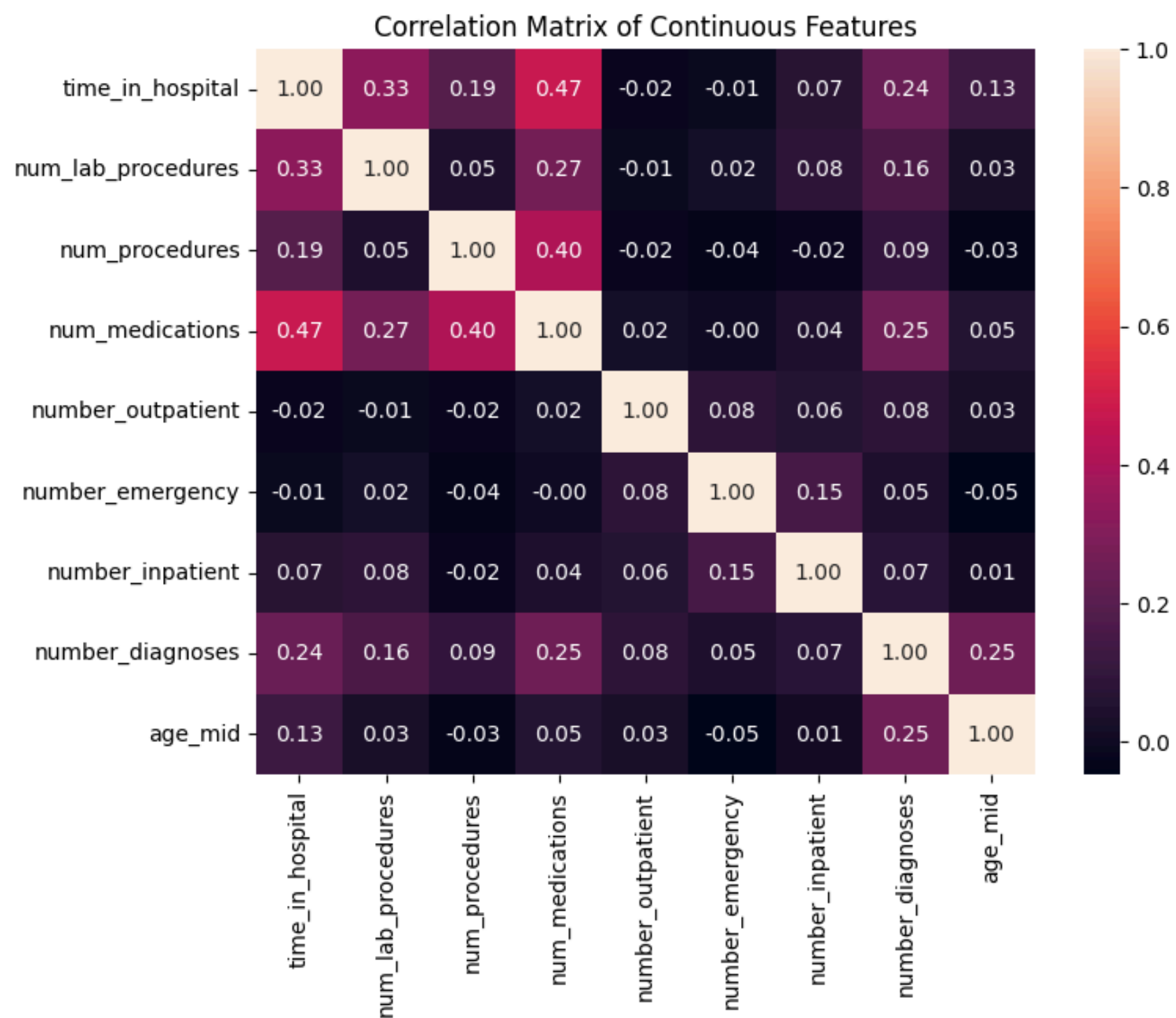
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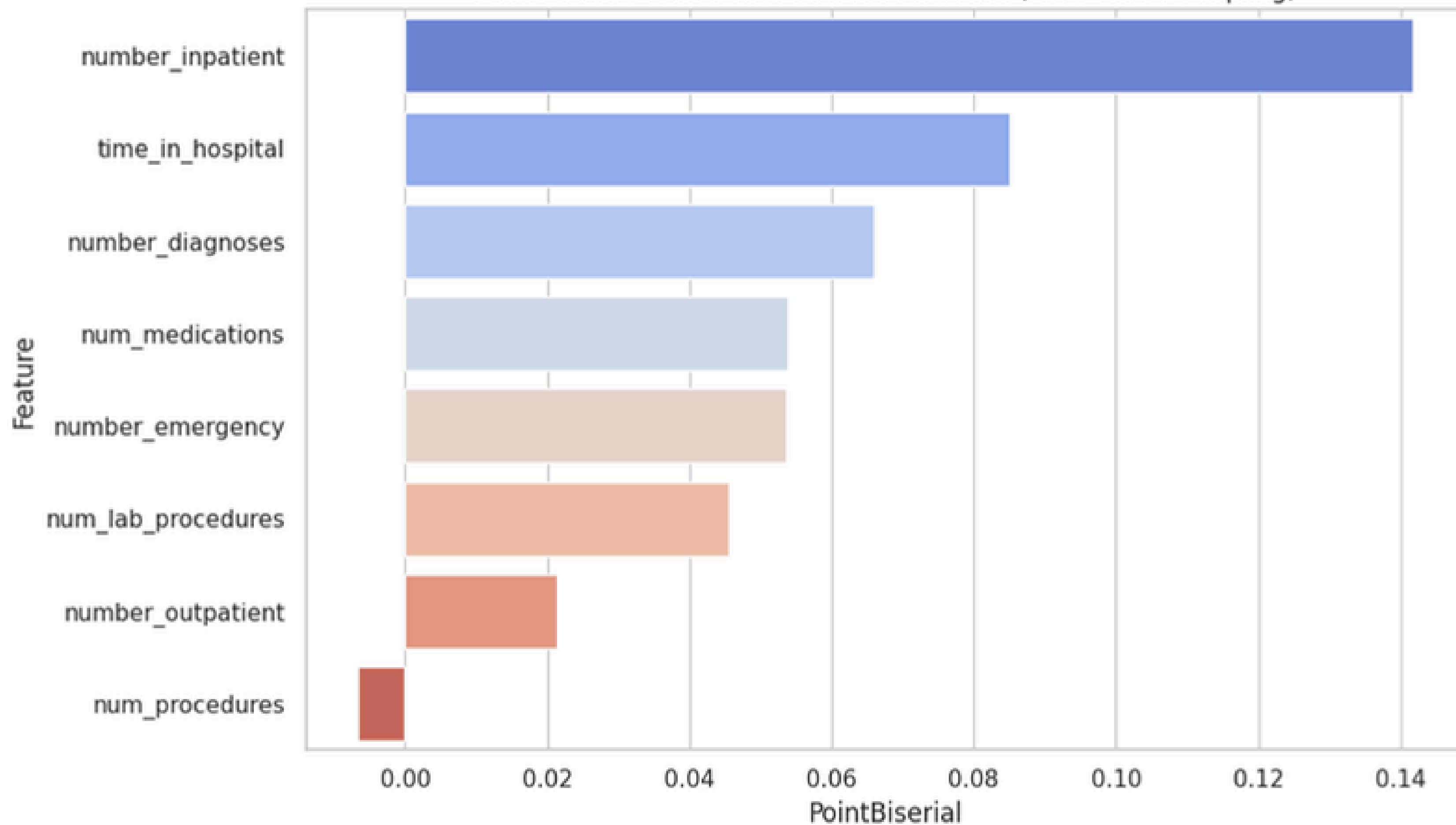


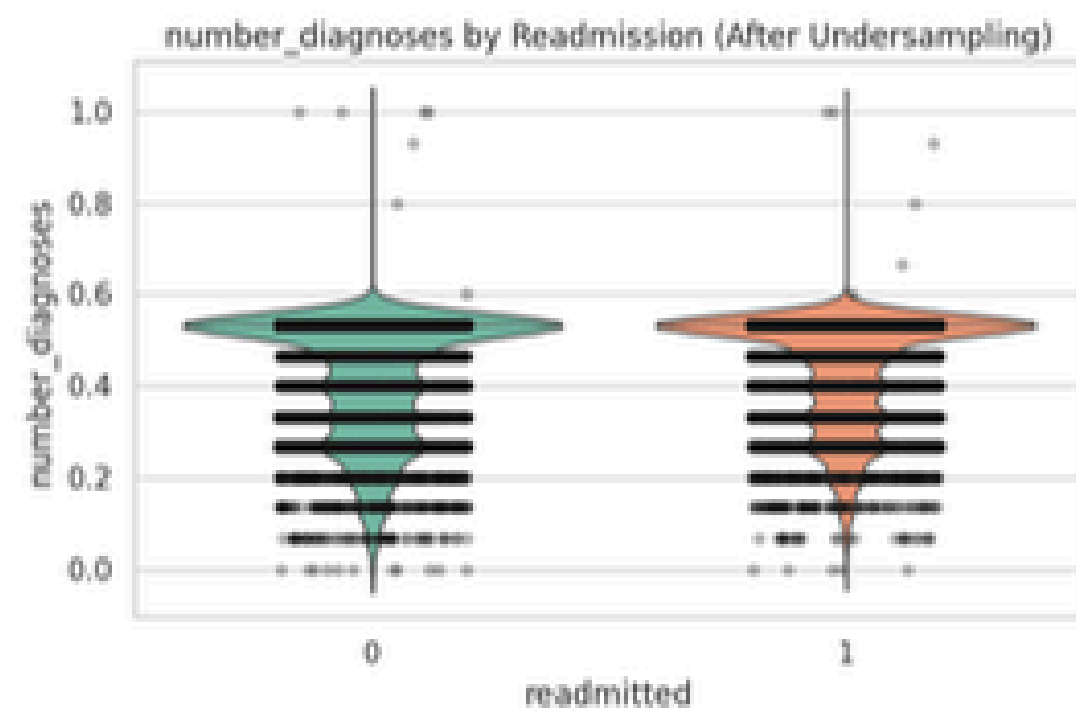
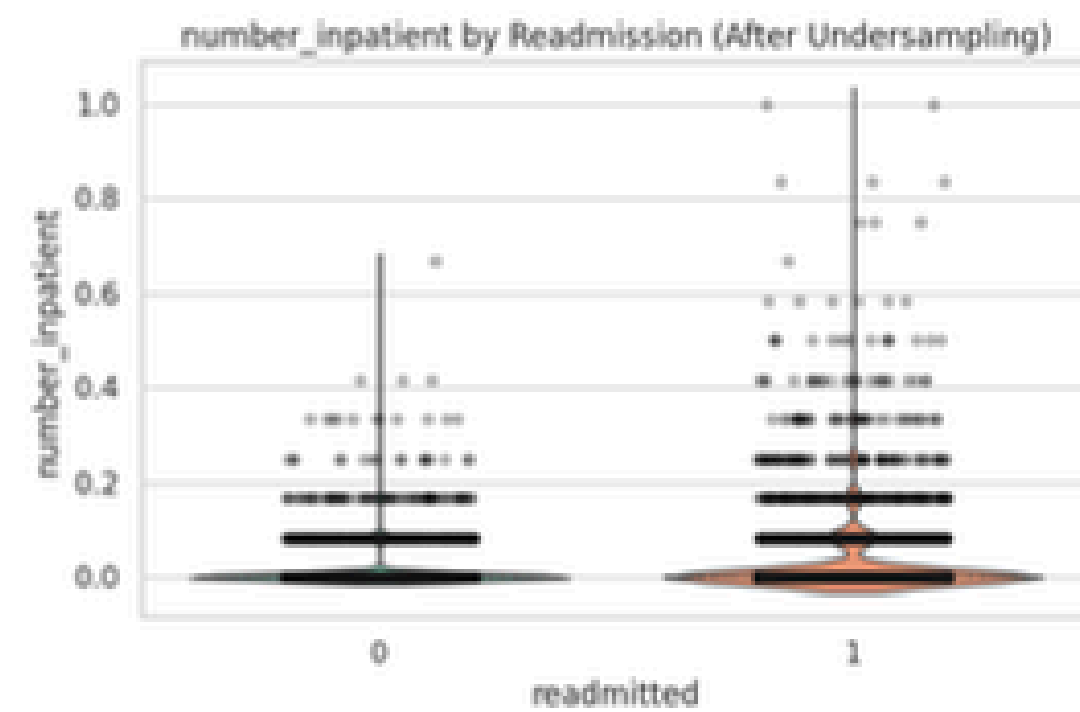
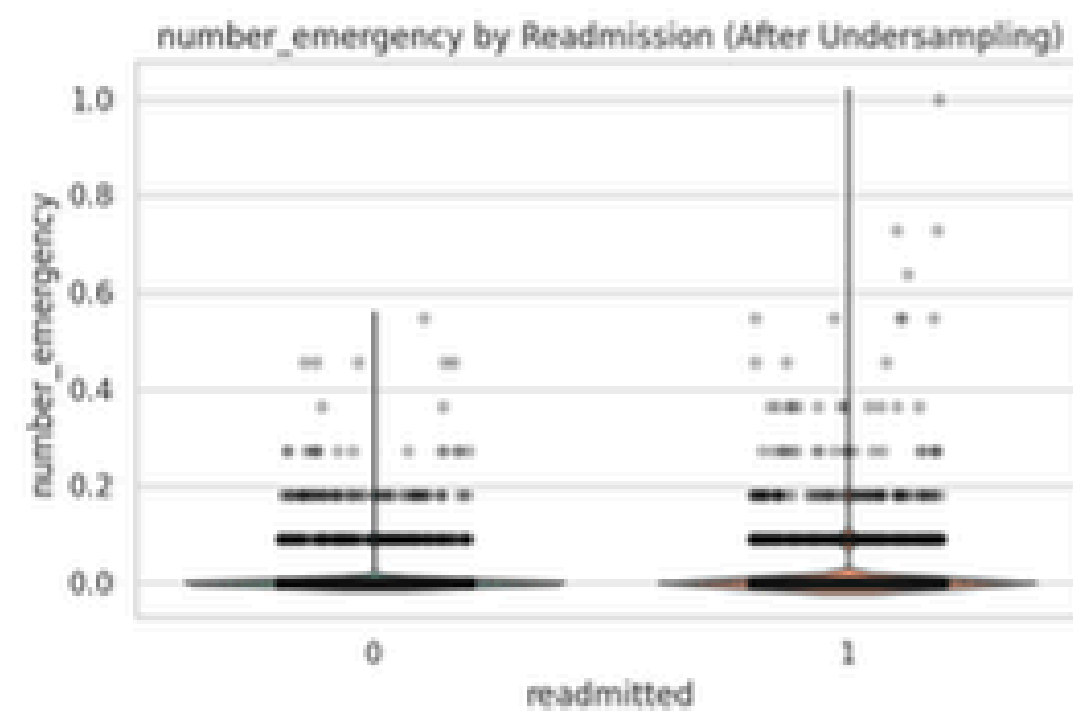
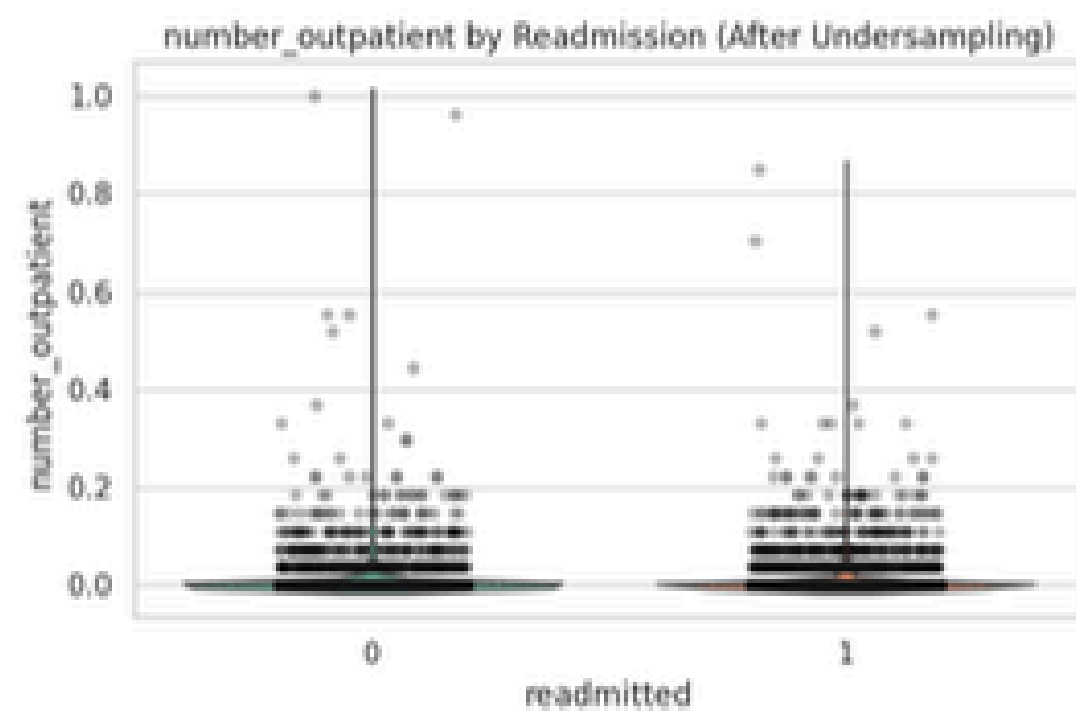
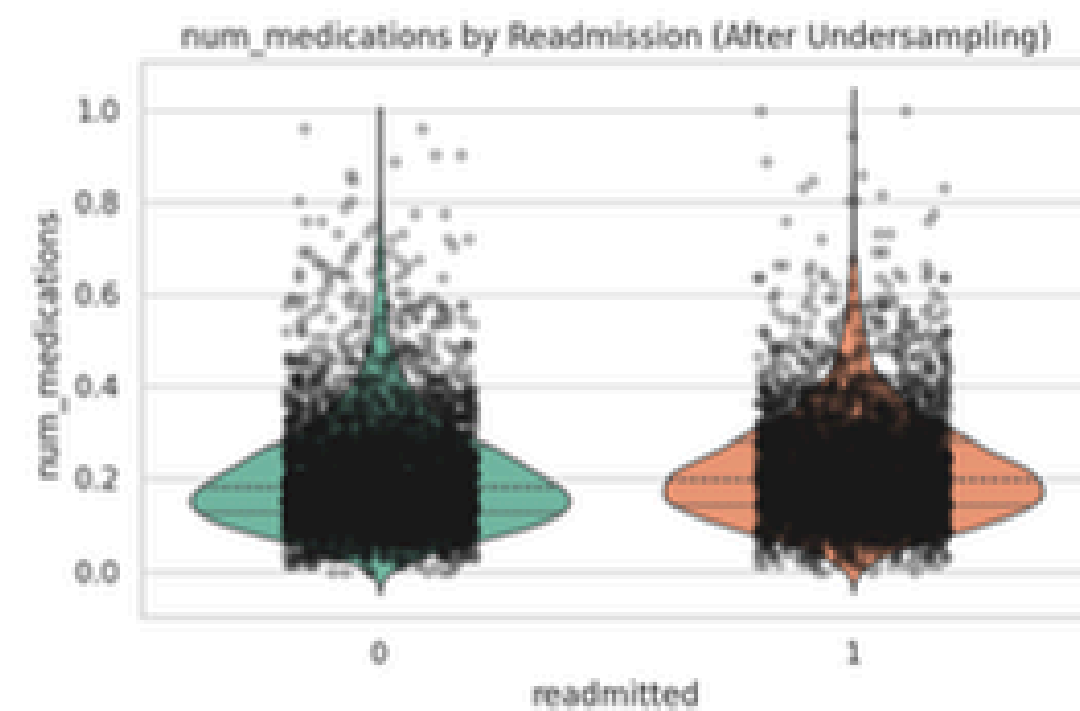
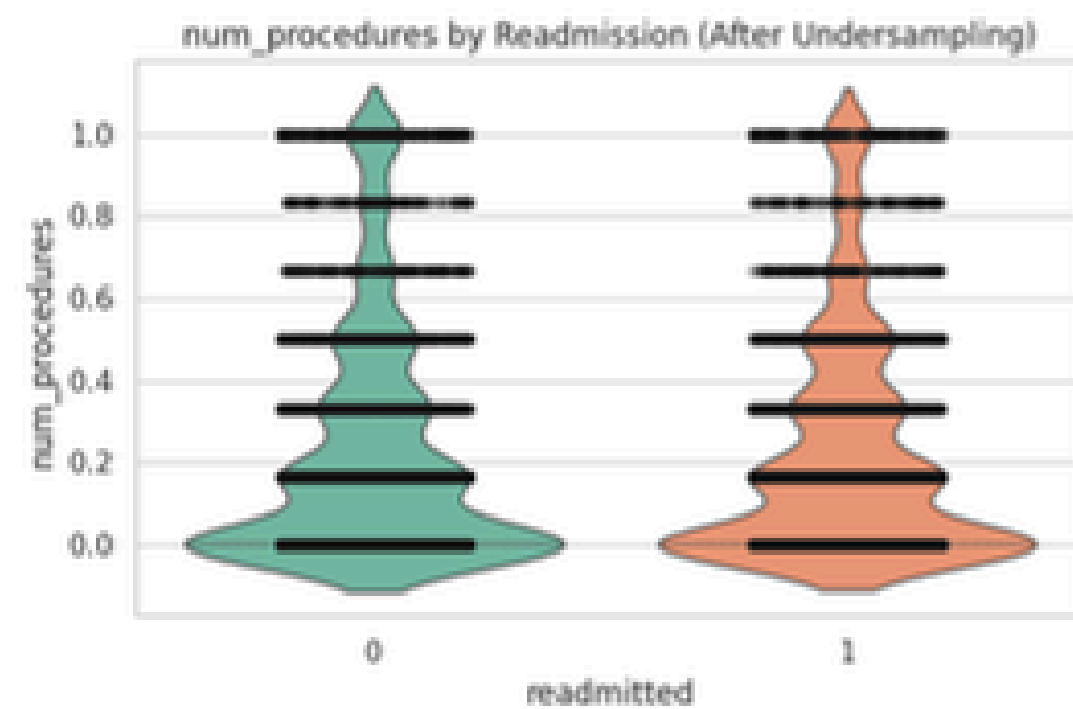
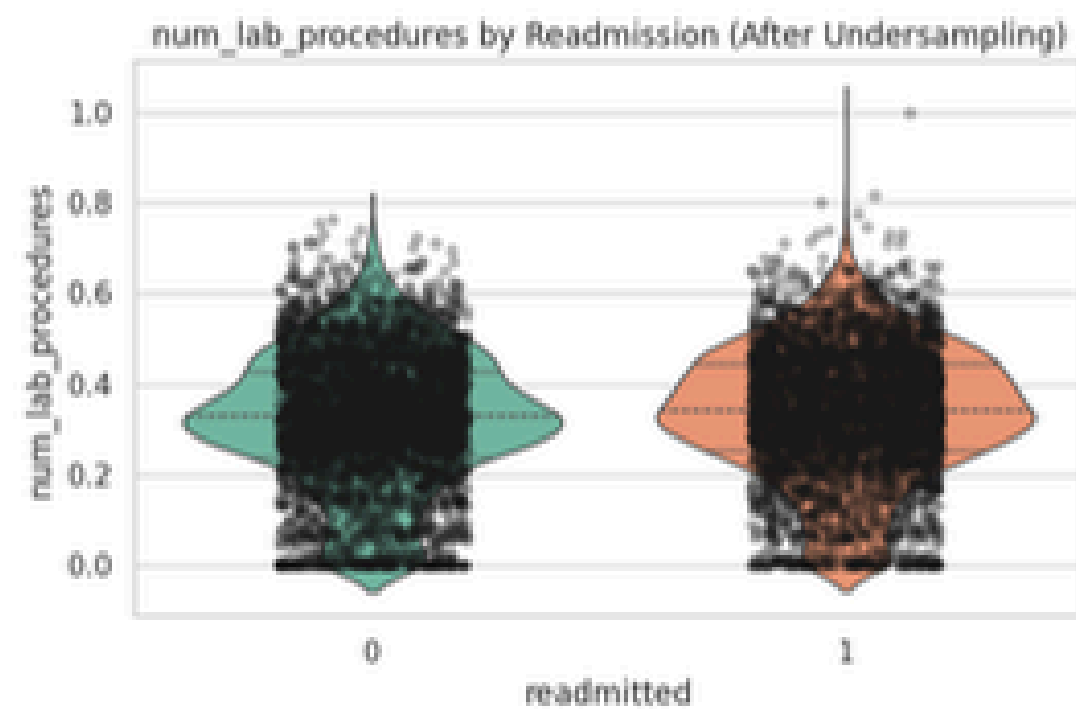
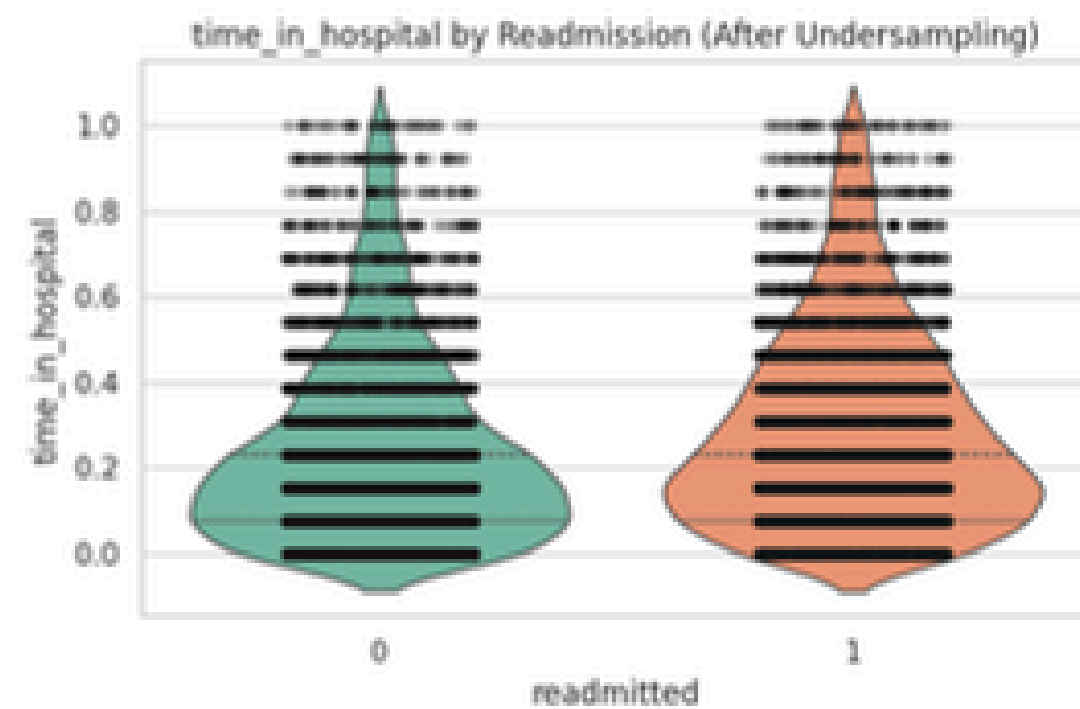
Exploratory Data Analysis + Pre-Processing Visualizations





Point-Biserial Correlation with Readmission (After Undersampling)





Methods

Goal: Optimize **F1 score**

ML CLASSIFIERS	WHY INCLUDE IT?
Logistic Regression (Baseline)	Establishes baseline F1 and accuracy; simple and transparent benchmark
Random Forest	Strong tabular model; good coparator to boosting and neural methods
XGBoost	Best-in-class benchmark for structure EHR data; expected higher F1
Transformer-Based Neural Network	Tests whether deep representation learning improves readmission prediction

Experiments



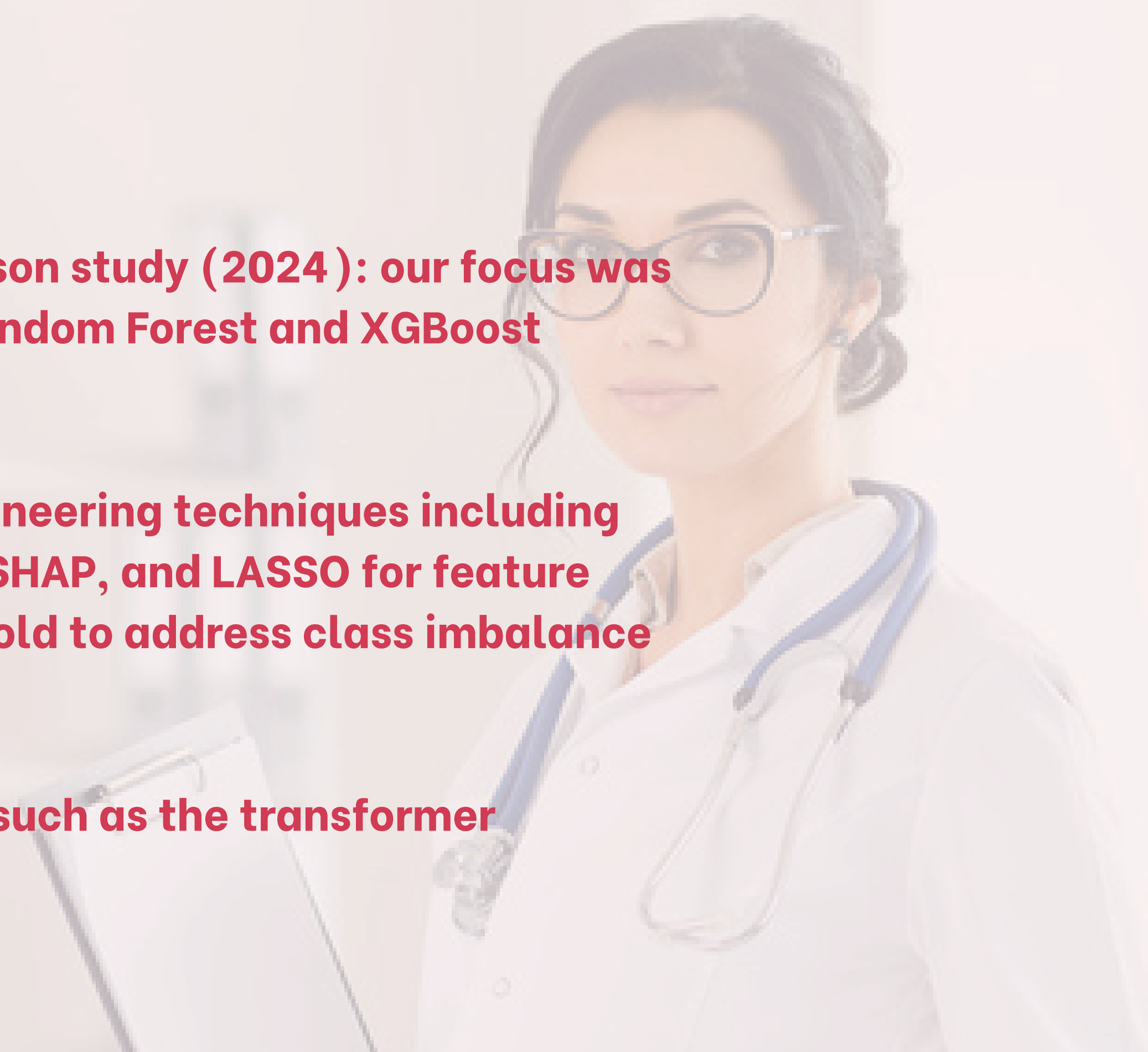
Expanded on Liu et al's comparison study (2024): our focus was on hyperparameter tuning on Random Forest and XGBoost



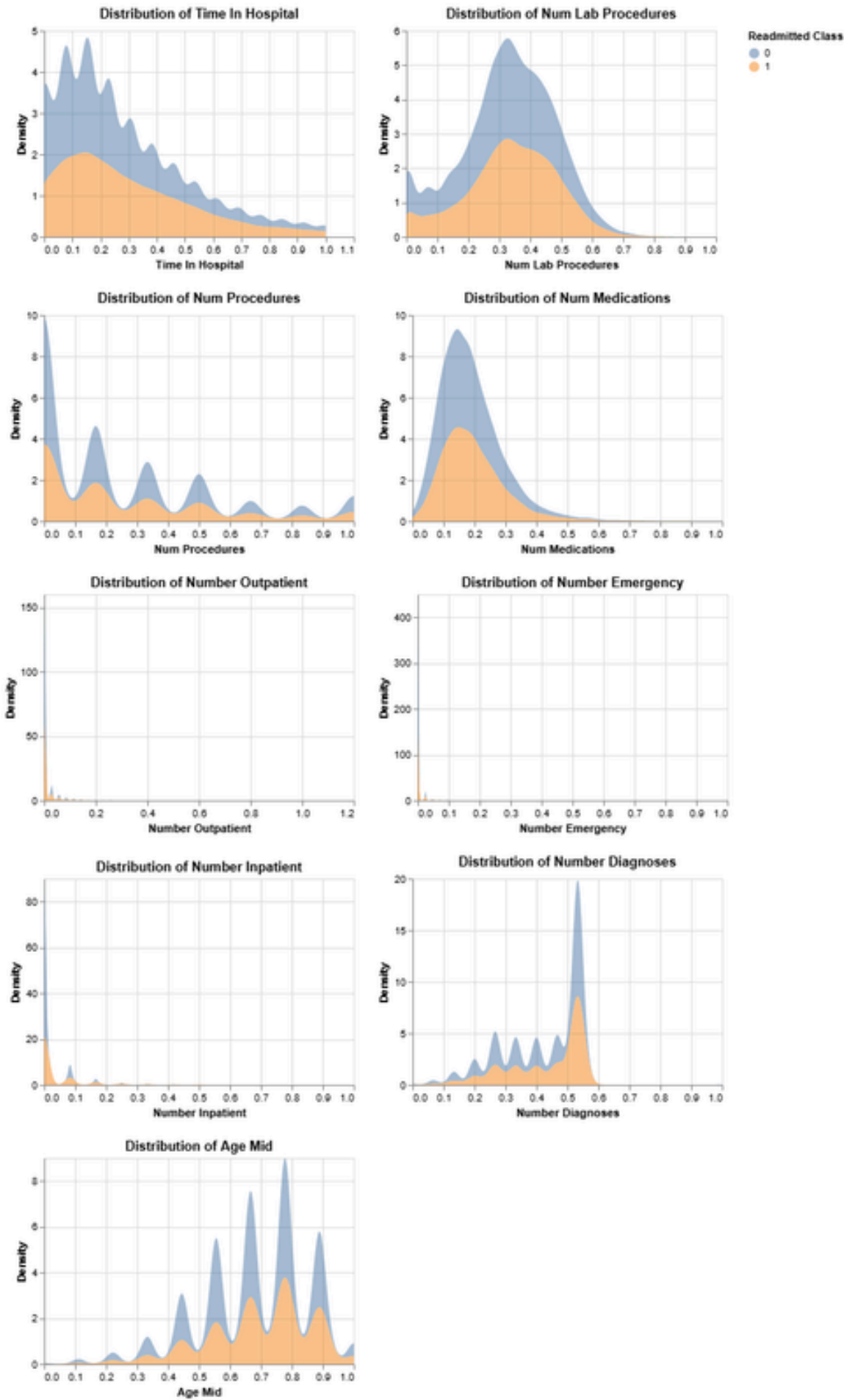
Evaluated advanced feature engineering techniques including Grey Wolf Optimization (GWO), SHAP, and LASSO for feature selection and SMOTE /Group K-Fold to address class imbalance



Explored Deep Learning Models such as the transformer architecture



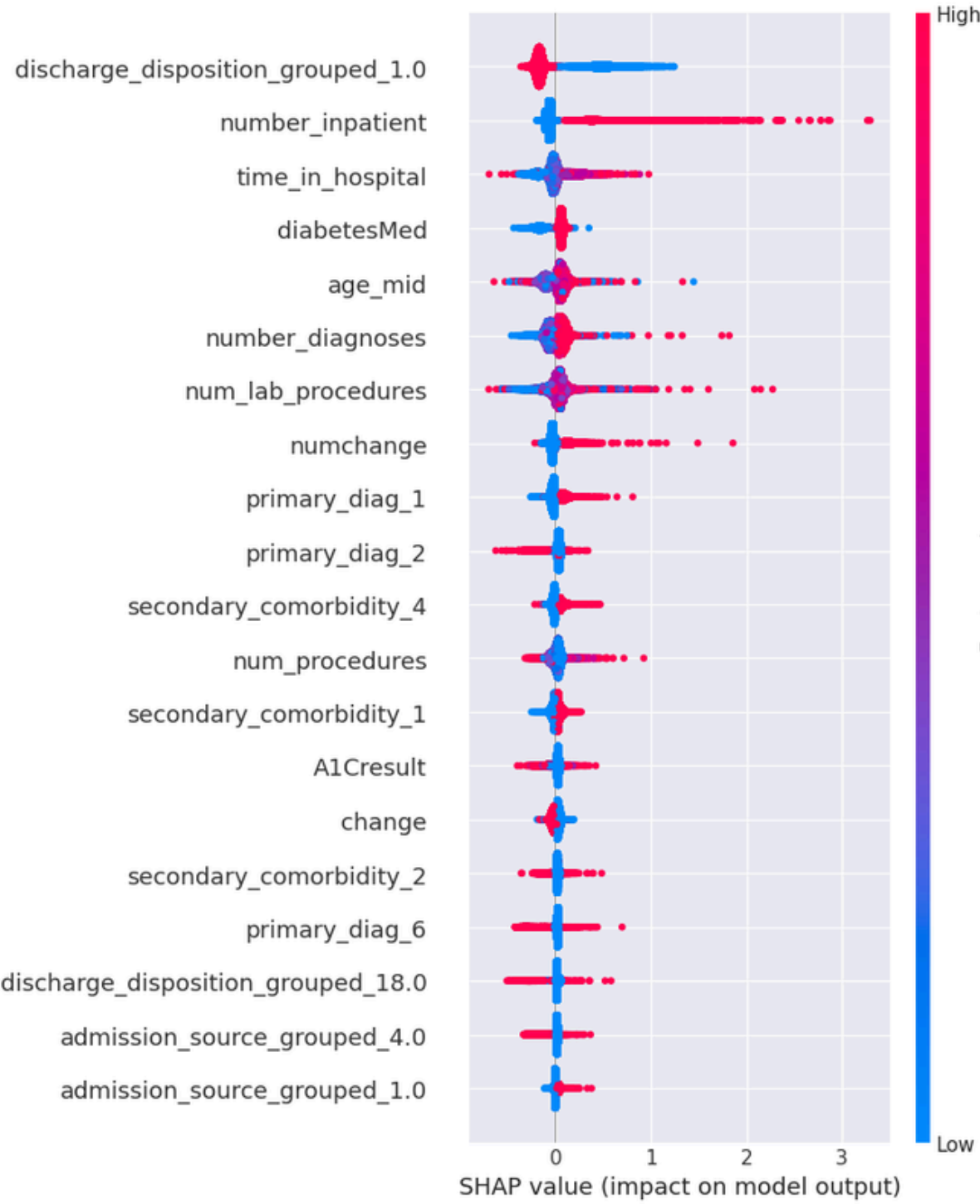
Raw Data



Power Transformations



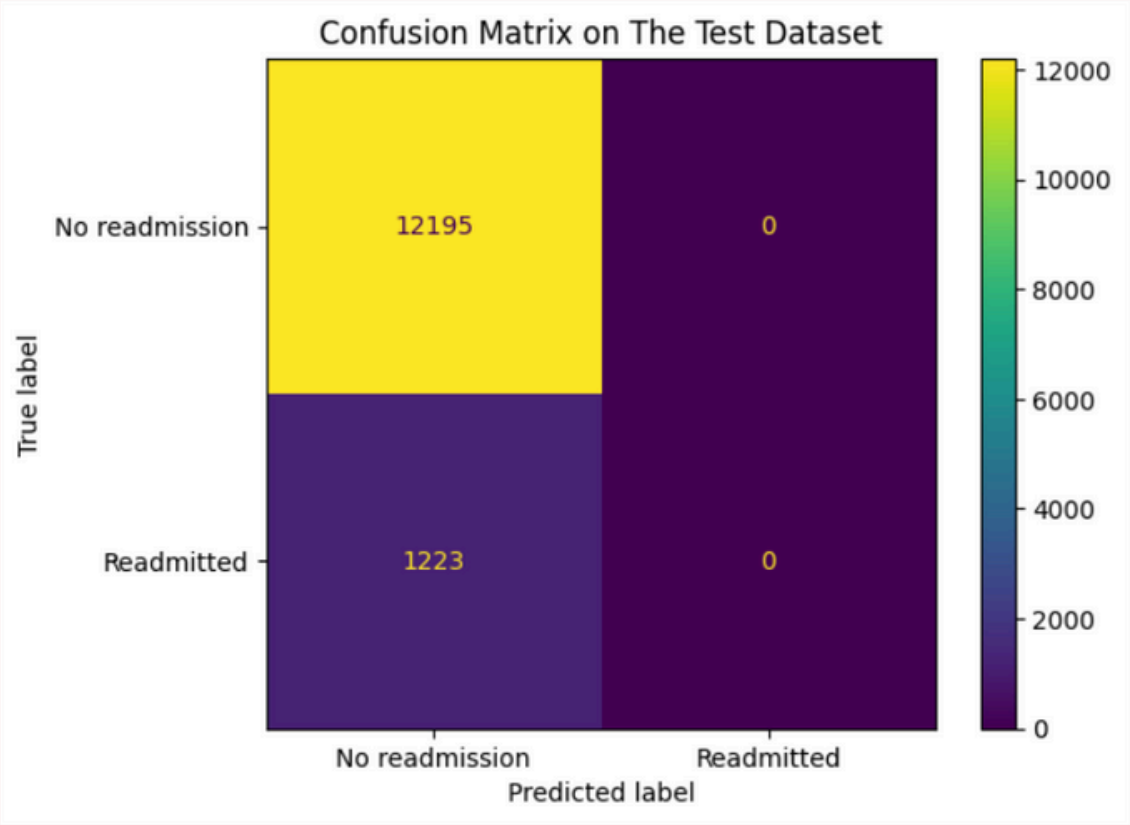
Shap Value Feature Selection



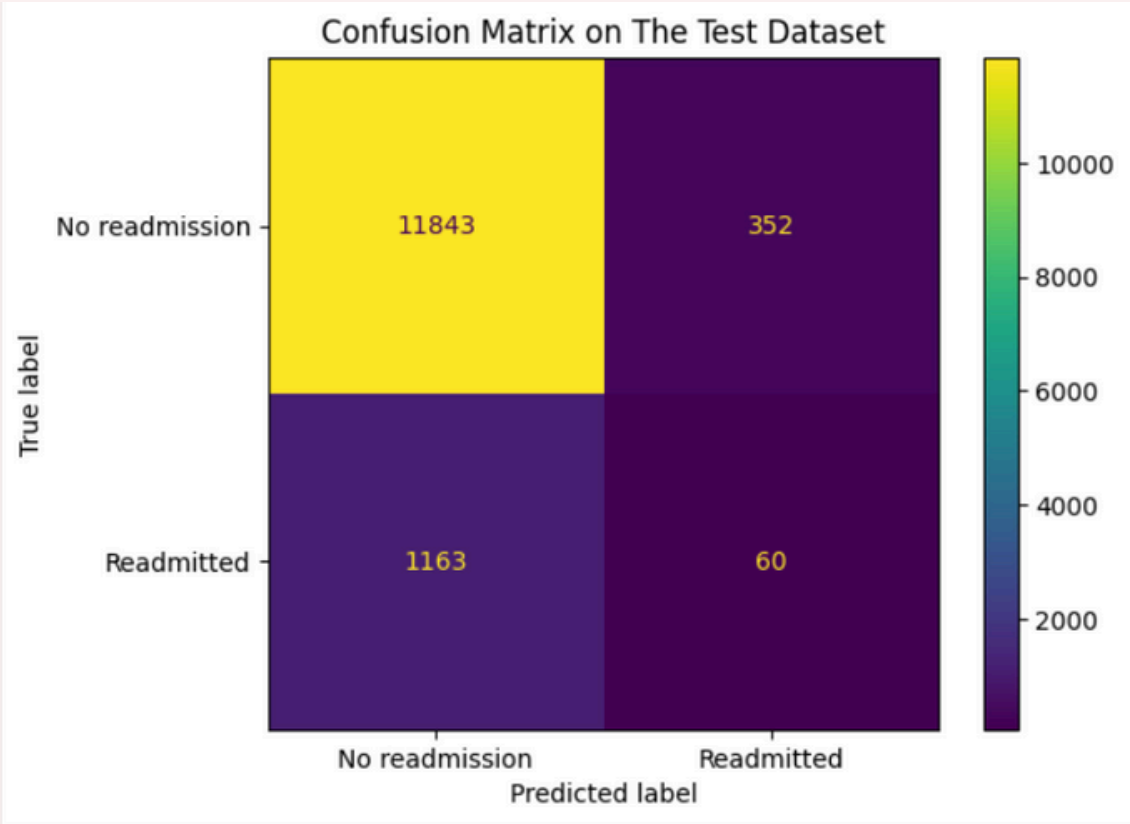
Results (Imbalanced Classes)

Classifier	Test Accuracy	F1 Score
Logistic Regression (Baseline)	90.88%	0%
Random Forest	90.89%	0%
XGBoost	88.71%	7.34%
Transformer-Based Model	90.89%	0%

Random Forest Model:
failed to identify the true readmitted cases due to class imbalance



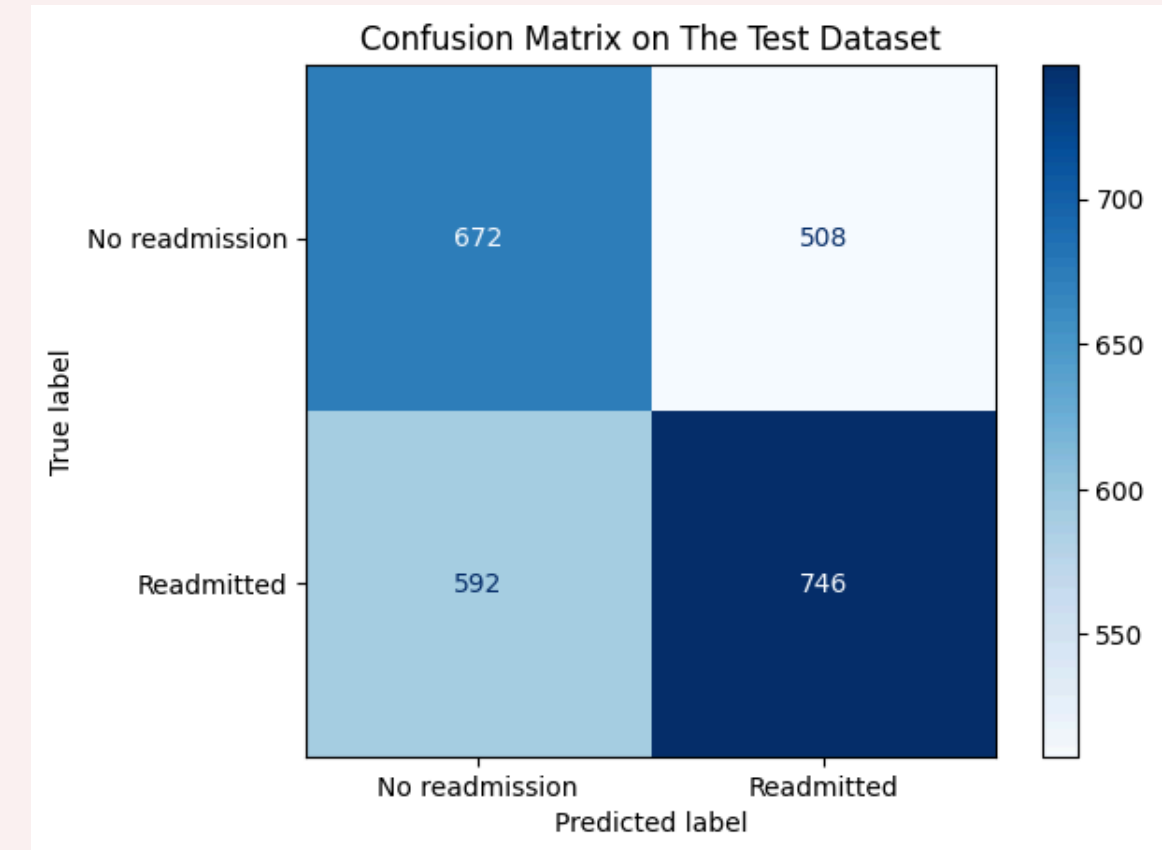
XGBoost Model:
able to identify some readmitted cases



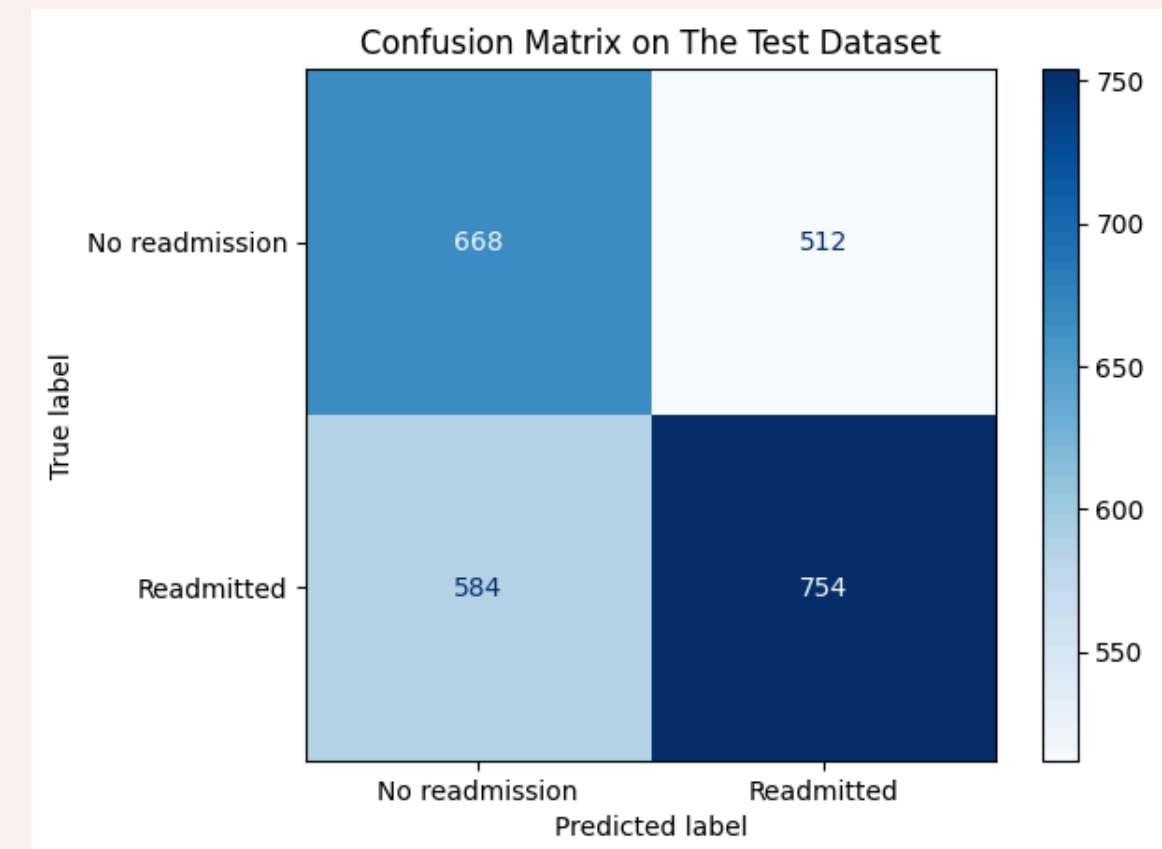
Results (Balanced Classes)

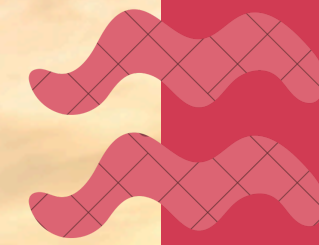
Classifier	Train Accuracy	Test Accuracy	F1 Score
Logistic Regression (Baseline)	53.60%	53.26%	52.60%
Random Forest	81.61%	56.31%	57.56%
XGBoost	60.35%	56.47%	57.91%
Transformer-Based Model	55.94%	52.70%	46.04%

Random Forest Model



XGBoost Model





Conclusions



- **ML can predict diabetic patient readmission rates, but the choice of model is crucial**
 - XGB model F1 score: 57.91%
- **A patient's overall medical history and complexity is the primary drivers of readmission**



Thank you!



Appendix

Feature Importance

Classifier	Importance Feature(s)
Random Forest	num_lab_procedures num_medications time_in_hospital
XGBoost	number_inpatient time_in_hospital diabetesMed

current patient complexity

prior health care utilization

Subgroup Analysis

Classifier	F1 Score	F1 Score - Female	F1 Score - Male
Logistic Regression (Baseline)	52.60%	50.29%	54.85%
Random Forest	57.56%	59.72%	54.99%
XGBoost	57.91%	59.81%	55.76%